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**How Much Can One Tile Be Worth?
Scrabble with a Fixed Number of Tiles**

Anonymous research manuscript

Exact move scores for one through seven tiles in NWL23 and CSW24.

How Much Can One Tile Be Worth? Scrabble with a Fixed Number of Tiles

Abstract. How much can a Scrabble player achieve with just one tile? The answer is 213 points in the North American dictionary, and 225 in Collins. Two tiles can score 459 and 483. We determine the maxima for every rack contribution from one through seven tiles, culminating in 1786 for NWL23 and 1787 for CSW24. The mathematics separates three easily confused objects: a collection of scoring words, a legal board, and a position that can actually occur in a game. Elementary deletion and exchange arguments reduce the upper bounds to finite, reproducible integer calculations. The last North American case becomes a small connectivity problem; the Collins construction needs a two-letter bridge. Comparing dictionaries reveals that the crucial extra word need not even be one of the words scored. The larger dictionary can also give the smaller maximum: 1721 rather than 1723 with six tiles. Independent forward verifiers check complete legal histories, permanent blanks, and both players' racks and draws.

Keywords: Scrabble, extremal combinatorics, computer-assisted proof, dictionary comparison.

1. TWO TILES, ONE HUNDRED AND FIVE POINTS There are Scrabble moves whose beauty lies in how little they add. In the broadcast position that motivated this investigation, Evan Chester places just a Z and an S, turning an existing AG into ZAGS. The Z lands on a double-letter square; the S, on a triple-word square, also completes VIDIOTS. The arithmetic is

$$3(20 + 1 + 2 + 1) + 3(4 + 1 + 2 + 1 + 1 + 1 + 1) = 72 + 33 = 105.$$

The supplied image identifies the event as the 2026 World Cup, round 17, against Steve Grob. We can check the score directly from that image; we do not use its reported two-tile record status as an independently established historical fact.

For documented tournament context, NASPA's records list CAZIKUES for 392, played by Karl Khoshnaw in Manchester on 11 April 1982 under OSW, and three 311-point plays in its North American tournament category. Its 365-point QUIXOTRY, by Michael Cresta in 2006, belongs to the separate club category [1]. These are achievements against actual opponents, under the dictionaries of their time. They are not upper bounds over all possible positions.

That second question has a long recreational history. Scrabulizer's account records constructed 1780-point TWL06 and 1785-point SOWPODS moves in 2008 [2]. Popular retellings circulated 1778 as a theoretical maximum [3]; Puzzling juxtaposes records under different dictionaries [4]. In 2024, Bob Lucassen found the 1786-point NWL23 construction reproduced here. Woogles reports that his satisfiability computation proves optimality [5]. We do not claim that construction as new, nor use the reported solver conclusion as our upper-bound certificate. Instead, we expose and independently reproduce the finite reductions. Even "possible" needs care: does it describe scoring words, a legal board, or a board that could arise in a game?

Nor is the Collins score 1787 new: the contributor DejMar reported it in 2016 and clarified in 2018 that the construction used CSW15, not the older SOWPODS list [6].

Our task is to certify the version-specific maxima and their attaining histories, rather than claim priority for previously reported numbers.

We begin at the small end. Let $M_k(D)$ be the largest ordinary move score obtainable by placing exactly k tiles in a game with dictionary D . How much of the spectacular score is already available when $k = 1$ or 2 ? The answers are surprisingly large, and changing dictionary changes the reason as well as the number.

2. THE RULES THAT MATTER The board has 15×15 squares. Columns are A–O, rows 1–15; H8 is the centre. There are 98 lettered tiles and two blanks. The first word covers H8. A later move places between one and seven tiles in one row or column, with no gap in the resulting word; existing tiles may fill its interior. The board remains connected. Every new maximal horizontal or vertical string of length at least two must belong to the dictionary [7].

Only newly covered premiums count. Letter premiums are applied before word premiums, and multiple word premiums multiply. A new premium contributes to each word formed through that square. Blanks have permanent value zero, though their squares still activate word premiums. Exactly seven newly played tiles earn an additional 50 points [7]. We exclude challenge bonuses, time penalties, endgame rack transfers, and unchallenged invalid words. The players' choices and draws may be extraordinarily cooperative; the definition is existential, not probabilistic.

The dictionaries are versioned NWL23 and CSW24 files, identified by hashes in the reproducibility instructions; licensed dictionary files are not to be redistributed with the public code. NASPA identifies NWL2023 as its competitive lexicon effective 29 February 2024 [8]; WESPA publishes CSW24 material and specifies CSW24 for its 2026 Open [9, 10]. The KWG files contain 212,868 and 299,162 entries respectively, but only 196,601 and 280,887 have board-playable lengths 2–15. Neither full-file count should be confused with its playable count.

Theorem 1 (The first two rows). *For the standard English board, tile supply, and scoring rules,*

	NWL23	CSW24
M_1	213	225
M_2	459	483

The maxima are attained on legally constructed boards.

The remainder of the proof has two different jobs. An upper bound may grant a fictitious player extra freedom. An attaining construction must obey every rule. Confusing these jobs was the source of an error in an earlier six-tile calculation.

3. FROM WHOLE GAMES TO A FEW INTERSECTING WORDS For $k \geq 2$, rotate the description by transposition if necessary so that the move is horizontal. The premium board is invariant under transposition. There is one main word W , containing every new square, and at most k perpendicular words, one through each new square. For $k = 1$, there are at most two words; either orientation may be absent. Words elsewhere contribute nothing to this move.

Lemma 2 (Deletion fragments). *Remove the newly played tiles from a scored word. Every remaining consecutive fragment of length at least two must belong to D .*

Proof. Those letters were already present. Each internal fragment ends at a square that was empty before the move. An exterior fragment cannot have a further adjacent

old letter beyond the scored word, since the scored word was chosen maximal after the move. Thus each nontrivial fragment was a maximal word before the move. A one-letter fragment need not itself be a dictionary entry: it may have been connected perpendicularly. Empty fragments impose no condition. ■

That final sentence matters in code. A new tile at the last position leaves an empty suffix, which is legal. Rejecting it silently deletes a family of possible plays. The reproduction tests explicitly include this boundary case for each $k = 2, \dots, 7$.

Lemma 3 (The premium ceiling). *The main-word multiplier of a k -tile move is at most $3^{\min(k,3)}$. Each perpendicular word has multiplier at most three.*

Proof. An edge row contains three triple-word squares and no double-word squares. The middle row has two triple-word squares and one double-word square, whose product is 18. Every other row has at most two double-word squares and no triple-word squares. Activating at most k of these premiums gives the stated ceiling; transposition covers columns. A perpendicular word contains only one new square, so activates at most one word premium. Old premiums never reactivate. ■

The ceiling concerns multipliers, not total scores: letter values, letter premiums, and simultaneous crosses still have to be optimized.

Write $v(a)$ for the face value of letter a , and λ_p, μ_p for the letter and word multipliers at square p . If I indexes the new letters of W , its score with real tiles everywhere is

$$A(W, I) = \left(\prod_{i \in I} \mu_{p_i} \right) \left[\sum_{i=1}^{|W|} v(W_i) + \sum_{i \in I} (\lambda_{p_i} - 1) v(W_i) \right]. \quad (1)$$

Old premiums are absent from this formula. A cross through new letter x at p scores $\mu_p[v(C) + (\lambda_p - 1)v(x)]$ before blanks.

For a fitted main placement P , let $C_j(P)$ be the largest score of any deletion-valid cross fitting through its j th new tile, including the zero option of no cross. Then

$$U(P) = A(P) + \sum_{j=1}^k C_j(P) + 501_{\{k=7\}} \quad (2)$$

is an upper bound. Each actual cross is in its enumerated domain; replacing it by the independent maximum can only increase the sum. Tile scarcity, word interactions, support tiles, connectivity, and playability have all been relaxed. Consequently the bound needs none of those assumptions to be true of its fictitious maximizer.

4. BLANKS BELONG TO SQUARES, NOT TO WORDS A tempting collection of words often uses two Z's or too many P's. The supply must be counted on the union of physical squares: a crossing tile is used once, although it contributes to two word scores.

For each occupied scoring square s , add its coefficients from every scored word to obtain c_s . Let n_a be the number of squares showing a , B_a the number of real a -tiles, and $e_a = (n_a - B_a)_+$. Order the coefficients of the a -squares as $c_{a,1} \leq \dots \leq c_{a,n_a}$.

Lemma 4 (Exact blank penalty). *The scoring pattern is impossible if $\sum_a e_a > 2$. Otherwise its maximum score under the physical tile supply, before any additional support is considered, is*

$$A_{\text{all}} - \sum_a v(a) \sum_{j=1}^{e_a} c_{a,j}. \quad (3)$$

The bingo bonus, when present, is added afterward.

Proof. At least e_a of the a -squares must be blanks. A blank at s loses exactly $v(a)c_s$. If a proposed assignment makes a larger-coefficient square blank while leaving a smaller-coefficient square real, swapping their physical tiles cannot lower the score. Repetition gives the displayed minimum loss. Letters have independent real-tile supplies; their only shared constraint is the total of two blanks. Using an additional, unforced blank cannot improve the score. ■

The lemma is exact for the scoring pattern and optimistic for its completion. Additional support tiles might force further blanks. For pruning a partially selected product, the same calculation is a lower bound on its eventual loss: a subset already exceeding a letter's supply must contain the excess blanks itself, and later crosses can only increase its coefficients.

For a one-tile move the formula simplifies. Every old square has coefficient μ_p , and the new square has coefficient $\lambda_p\mu_p$ or $2\lambda_p\mu_p$. Since each English letter has a real tile, a maximum keeps the new tile real; swapping it with an old copy otherwise improves or preserves the score. Thus the penalty is $\mu_p \sum_a e_a v(a)$.

5. THE EXHAUSTIVE PART OF THE SMALL THEOREM For one tile, enumerate every dictionary word and distinguished letter occurrence passing Lemma 2. Fit it to each square in both directions. Include single-word moves, and pair horizontal and vertical occurrences with the same new letter. Apply Lemma 4. Lists are sorted by all-real score; a suffix is skipped only if even its all-real score is below the current maximum. Every omitted score is therefore bounded above by an already examined score. The maxima of this relaxed space are exactly 213 and 225.

For two tiles, enumerate every word of length 2–15, every pair of new indices, and all fitting horizontal placements. The enumeration is over all choices, not a list of plausible long words. Table 1 gives the freshly reproduced counts. At threshold 459, only 81 NWL23 placements and 256 CSW24 placements survive the independent bound. Enumerate their complete cross products, retaining the no-cross options, enforcing the common bag, and applying the exact blank penalty. No incidental-word rejection is used. NWL23 leaves one pattern at score at least 459, scoring 459; CSW24 leaves 369, none above 483. These are upper bounds on real play, regardless of whether any retained pattern is completable.

This establishes the four upper bounds in Theorem 1. The search uses integer arithmetic. A second, deliberately direct enumerator checks the common C++ scanner against every placement in a small artificial dictionary for $k = 2, \dots, 7$. Further adversarial tests check transposition, impossible three-blank demands, and a shared square whose blank penalty must be counted in two words. These tests support the implementation; the exhaustive dictionary runs, rather than the tests alone, supply the numerical part of the proof.

6. FOUR BOARDS THAT ATTAIN THE BOUNDS Figure 1 displays the Collins two-tile position. The supplementary board atlas displays all four small-case witnesses

Table 1. Fresh exhaustive counts for one and two tiles. The last three rows use the inclusive two-tile threshold 459.

Quantity	NWL23	CSW24
Playable dictionary entries	196,601	280,887
Raw distinguished occurrences	1,764,090	2,557,653
Deletion-valid occurrences	240,528	387,172
Raw pairs of new indices	7,656,225	11,273,568
Deletion-valid pairs	817,488	1,409,087
Fitted main placements	91,639,770	155,270,805
Independent-bound survivors	81	256
Bag-feasible products	1	369
Maximum product score	459	483

with their complete final racks. Its final section lists their placement histories, including scores of every setup move. The verifier checks board boundaries, alignment, word membership, connection to the centre, every incidental word, new-tile counts, permanent blanks, and the physical bag after every move.

For the NWL23 one-tile witness, add S at O15. This completes two words: HYDROXYZINES and MICROTECHNIQUES. Their face sums are 38 and 33. Each receives the new triple-word premium, giving $3(38 + 33) = 213$. The old dictionary words are HYDROXYZINE and MICROTECHNIQUE.

In CSW24, the same S completes METHOXYBENZENES and QUICKSILVERINGS, worth $3(41 + 34) = 225$. None of these two words or their singular predecessors occurs in our NWL23 input. Both one-tile constructions use no blanks.

For two tiles in NWL23, play Y at H15 and S at O15. These join PA and CHEQUE into PAYCHEQUES. The two triple-word squares multiply the main word by nine. The crosses are XYLOGRAPHY and KATZENJAMMERS, each tripled:

$$9 \cdot 29 + 3 \cdot 29 + 3 \cdot 37 = 261 + 87 + 111 = 459.$$

An auxiliary A at I13 is a blank, outside every word scored on the final move.

For CSW24, retain the same new letters and squares but complete JAYHAWKERS, QUALIFICATORY, and METHOXYBENZENES:

$$9 \cdot 30 + 3 \cdot 30 + 3 \cdot 41 = 270 + 90 + 123 = 483.$$

Its 11-move setup uses no blanks. One useful construction starts with ALIF through H8 and grows FIREBOMB across row 8. This provides a bridge to BENZENE in column O. A separate CAT below ALIF allows the gap to be filled while making QUALIFICATOR. The finished board uses just 48 tiles.

There is also a rack issue, which a bare sequence of board moves can conceal. For these four witnesses it has a simple resolution.

Lemma 5 (Realizing a sparse history). *Suppose a legal, positive-scoring placement history uses at most 86 physical tiles altogether, with at most seven newly placed on each move. It can be realized by two players with legal draws and intervening passes.*

Proof. Give the second player seven tiles not used anywhere in the history, and let that player pass throughout. List the first player's required tiles in chronological order of placement, choosing any order within each move. Append unused physical tiles until the list has seven more tiles than the whole history needs; at most $86 + 7 = 93$ tiles

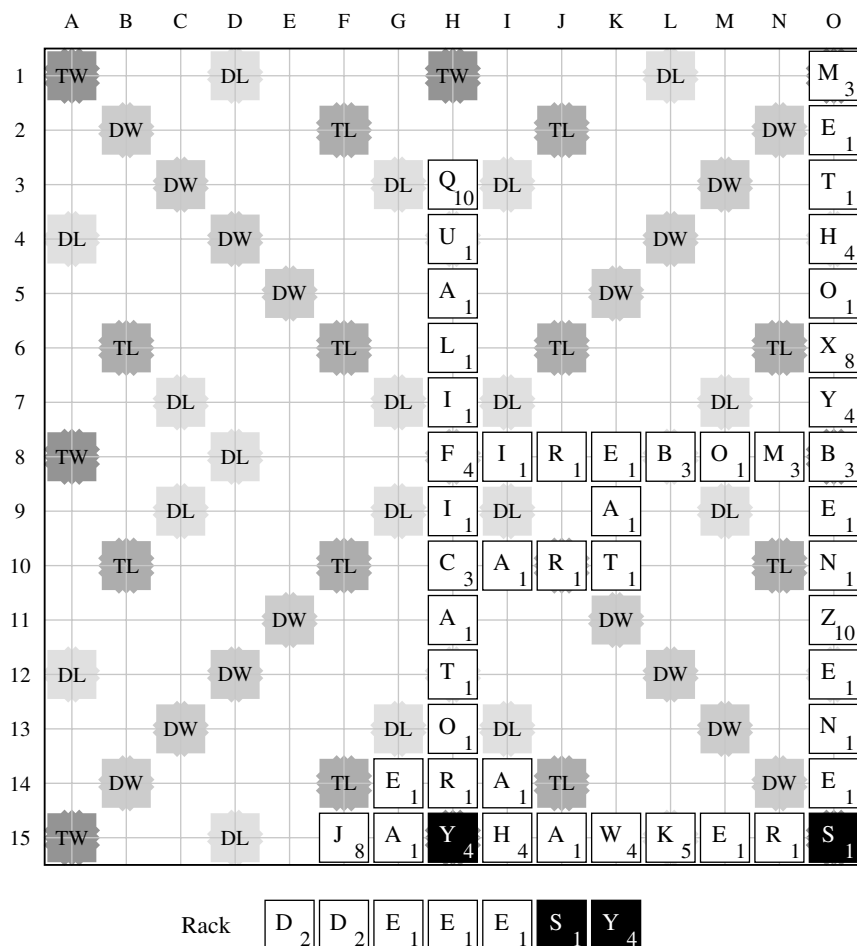


Figure 1. A 483-point two-tile move in CSW24. Black tiles are newly placed; the complete seven-tile rack appears below. Each premium is labeled, so the board remains readable in monochrome.

are required for this stream after the opponent's seven are reserved. Deal its first seven tiles and draw along the stream. Before every prescribed move the rack contains the next seven tiles of the stream, so it contains that move's at most seven required tiles. After playing them, draw the next tiles. Every prescribed move scores positively in the displayed certificates, so intervening single passes do not end the game by a scoreless-turn rule. ■

All four boards use at most 48 tiles. Lemma 5 therefore supplies legitimate racks and draws as well as legal board histories. This completes the attainment part and proves Theorem 1.

7. WHAT THE EXTRA DICTIONARY WORDS ACTUALLY CHANGE The twelve-point one-tile increase is transparent: the sum of the two word values rises from 71 to 75, and the new square triples the four-point gain. The two-tile increase of 24 has three contributions: nine from the main word, three from the first cross, and twelve from the second cross. They add to $9 + 3 + 12 = 24$.

But the vocabulary change is subtler than those score differences suggest. JAYHAWKERS and QUALIFICATORY are in both dictionaries. The CSW construc-

tion nevertheless cannot be imported into NWL23: JA, the fragment before the new Y, is absent there; so is QUALIFICATOR, which must precede the final Y in the cross. The other cross, METHOXYBENZENES, and its predecessor are also absent. Membership of the final words alone would miss two independent obstructions.

The dictionaries are not nested. The NWL23 word OPACIFICATIONS is absent from CSW24, yet is essential to our six- and seven-tile constructions. A larger dictionary can enable a different construction while disabling the one already found. Reusing the algorithm is appropriate; reusing either the numerical answer or its witness without checking every word is not.

8. THREE TILES: THE SAME MAIN WORD, A DIFFERENT ANSWER

Proposition 6. *The three-tile maximum in CSW24 is 912.*

Proof. At the inclusive threshold 883, the independent bound leaves seven main placements. Complete cross-product enumeration with the common physical bag, accelerated by Lemma 9 below, leaves 11,350 scoring products. Their largest value is 912. No support-word or connectivity assumption is used in this exclusion, so it bounds every genuine move.

To attain it, place W,S,S at A15, H15, O15 and form WHIPPERSNAPPERS. The old fragments are HIPPER and NAPPER. Two old P's, at D15 and E15, are blanks; the main word therefore scores $783 - 2(3 \cdot 27) = 621$. The crosses are KABELJOUW, METHOXYBENZENES, and QUALIFICATIVES. They score 75, 123, and 93, respectively. Thus the total is $621 + 75 + 123 + 93 = 912$.

Here is the complete setup, in order; a coordinate names the first square of a word, and H or V gives its direction:

1	V	BENZENE	H8	8	H	EUTECTIC	H9
2	V	METHOXYBENZENE	H1	9	V	CAT	O9
3	H	EE	G14	10	H	ORIENTAL	H5
4	H	HIPPER	B15	11	V	ALIF	O4
5	V	SH	B14	12	V	QUALIFICATIVE	O2
6	V	OU	A13	13	H	AE	N14
7	V	KABELJOU	A7	14	H	NAPPER	I15

Each step places at most seven tiles and scores positively. The verifier checks every word and bag count after every step, and the final board has 65 tiles. Lemma 5 supplies the racks and draws; their explicit stream is also included in the certificate. ■

The NWL23 three-tile construction uses the same main word, new squares, and main-word blank penalty. Its crosses score $60 + 105 + 96 = 261$; Collins increases these to $75 + 123 + 93 = 291$. The net gain is thirty points, with no contribution from the main word. The CSW-only terminal W in KABELJOUW is especially telling: KABELJOU itself is in NWL23, but the final hook is not. The short supporting words EE and OU are also Collins-only. Again, a dictionary matters both on and away from the scored words. The board and complete final rack appear in the supplementary atlas.

9. FOUR TILES: CHEMISTRY AND THE CENTRE SQUARE

Proposition 7. *The four-tile maximum in CSW24 is 1349.*

Proof. At threshold 1350, the complete independent scan leaves only three main placements from 953,946,180 fitted placements. The resource bound excludes one; exhaustive products exclude the other two even before checking board completion. This proves the upper bound.

Place M,B,Z,S at A15, H15, K15, O15. The main word is METHOXYBENZENES, scoring $27(41) = 1107$. Its four crosses are HYPOCHONDRIASM, JAMB, WOOTZ, and QUICKSILVERINGS. A2 holds an old blank H, reducing the first cross from 90 to 78. The total is

$$1107 + 78 + 45 + 17 + 102 = 1349.$$

The second blank is a supporting H at B14 and is not scored by the final move. The checked 21-move setup uses 72 tiles, so Lemma 5 applies. Its board, complete rack, and machine-readable history are in the supplement. ■

This construction has a convenient spine. Begin with TUTOR at H8, extend it to TUTORIAL, and grow TIRED down column J. The word JUDO connects it to both JAM and WOOT. On the right, SILVER grows into SILVERING, then QUICKSILVERING. On the left, FORESTER becomes DEFORESTER, providing the anchor for HOND, CHONDRI, and finally HYPOCHONDRIAS. Small hooks attach the bottom fragments ETHOXY, EN, and ENE.

One tied scoring pattern illustrates the difference between a legal diagram and a reachable game. Replacing two crossings by PROVERB and JEEZ also gives 1349 in the scoring relaxation. But PROVERB would run from H9 to H15. Its maximality forces H8 to be empty. Every game after the opening has a tile at H8, which can never be removed. No amount of support-word repair can save that pattern. We retained it in the upper-bound relaxation, but could not use it as an attaining witness.

The difference from the NWL23 optimum is 219 points. The main word changes from OXYHAEMOGLOBINS, scoring 891, to the Collins-only METHOXYBENZENES, scoring 1107. Of the gain, 216 points come from the main word and three from the aggregate cross contribution, which rises from 239 to 242. This is a different mechanism from the thirty-point three-tile gain, which comes entirely from crossings.

10. FIVE TILES: A MISSING CENTRE COSTS TWO POINTS

Proposition 8. *The five-tile maximum in CSW24 is 1478.*

Proof. The independent scan at threshold 1479 leaves twelve main placements among 1,552,255,935 fitted placements. Complete cross products under the common tile supply leave just ten patterns: three score 1480, seven score 1479. Every one has PROVERB from H9 to H15, so every one forces H8 empty. The opening covers H8 permanently, giving the required contradiction.

Place M,H,B,Z,S at A15, D15, H15, K15, O15. Again the main word is METHOXYBENZENES. The new H activates D15's double-letter square, while the three triple-word premiums still multiply. The main score rises from 1107 to $27(41 + 4) = 1215$, giving

$$1215 + 78 + 21 + 45 + 17 + 102 = 1478.$$

The crossings, in order, are HYPOCHONDRIASM, PENNYWORTH, JAMB, WOOTZ, and QUICKSILVERINGS. A2 is a blank H, costing twelve points; D10 is a blank Y in PENNYWORTH, costing four. All new tiles are real. The independent verifier checks the 24-move setup and its 81-tile final board; Lemma 5 again supplies racks and draws. ■

The construction shares its right-hand spine with the four-tile board. A YIELDER across D10–J10 provides the blank Y. Build PENNY on that anchor and extend it to

PENNYWORT. The bridge CLAP links its top to CHONDRI; extend this to the required left-hand word. The old bottom fragments are now ET, OXY, EN, and ENE. Five new tiles simultaneously complete six words.

The extra new H adds 108 points to the main word and 21 through its cross, accounting for the 129-point jump from four tiles. Compared with NWL23's 1401, however, the main-word advantage is 81 and the total cross contribution is four points lower. The net dictionary advantage is $81 - 4 = 77$, much smaller than the four-tile advantage of 219.

11. THE LARGER NWL23 CASES AND A REPAIRED PROOF The common scanner has been rerun for $k = 3, \dots, 7$. Table 2 reports its complete counts at the first score above the earlier attaining constructions. For seven tiles the threshold 1737 is before the bingo; it corresponds to a total of 1787.

Table 2. NWL23 independent-cross scans. Thresholds are pre-bingo scores. These counts alone do not establish optimality.

k	Threshold	Raw index sets	Fitted placements	Survivors
3	883	21,441,146	263,013,720	4
4	1131	43,492,783	558,818,205	10
5	1402	67,766,643	907,615,980	2
6	1724	83,642,940	1,157,520,345	9
7	1737	82,937,692	1,187,919,660	87

For three through five tiles, fresh exact product exclusions remove every survivor even before asking whether its supporting board is legal. Attainment illustrates three distinct uses of the same triple-triple-triple geometry. The main word runs from A15 to O15 in each case.

With three tiles, place W, S, S at A15, H15, O15 to make WHIPPERSNAPPERS. Its four P's require two blanks, at E15 and K15 in this construction. The resulting main score is $27(29 - 6) = 621$, not the all-real 783. The crosses RICKSHAW, OBJECTIVIZES, and DEMISEMIQUAVERS contribute $60 + 105 + 96$, giving 882. A twelve-move setup grows OBJECT through the centre into OBJECTIVIZE; VENDETTA connects it to the right-hand cross, and RABIETIC to the left. The two bottom fragments are HIPPER and NAPPER.

With four tiles, place O, Y, O, S at A15, C15, H15, O15 to make OXYHAEMOGLOBINS. The main word contributes 891. Its crosses, BALDACHINO, PLETHYSMOGRAPHY, JOCKO, and VENTRILOQUIZES, contribute $54 + 26 + 54 + 105$, totaling 1130. The blanks are both H's in the ordinary column-C cross, at C5 and C14. Their combined loss is eight points. Putting either in the tripled main word instead would be much more expensive. The 25-move setup leaves X, HAEM, and GLOBIN on the bottom row; the isolated X is supported by AX.

With five tiles, the main word is DICHLOROBENZENE. Its new letters are D, L, O, Z, E, at A15, E15, H15, L15, O15. No blanks are needed. L15 doubles the new Z, so the main word scores $27(32 + 10) = 1134$. The crosses HYPERVENTILATED, OVERFULFILL, JOCKO, QUARTZ, and MYXAMOEBAE contribute $81 + 20 + 54 + 34 + 78 = 267$, giving 1401. In the 21-move setup, NAPERIES through H8 anchors two left-hand crosses; FADO and CITER carry the construction to the right. The old bottom fragments are ICH, OR, BEN, and EN.

The three final boards have 61, 84, and 78 tiles. Complete histories and permanent blank assignments are checked in the certificates, and Lemma 5 supplies playable rack streams. The supplementary atlas shows every board and its complete final rack.

Six tiles need a stronger argument. The nine independent-bound survivors produce exactly twenty bag-feasible scoring patterns at 1724 or more, including apparent 1725s. An earlier program rejected them because the union of their scoring words already contained an illegal incidental string. That rejection is unsound: extra support tiles can sometimes repair such a string. For example, the forced row fragments C.NQ and C.EQ can occur inside CINC and CHEQUE, respectively. Dots denote still-optional support, not permanently empty squares.

The corrected test asks whether a row can be completed at all. Enumerate every dictionary word on every interval matching the forced letters and avoiding forced-empty cells. Permit isolated letters too. A left-to-right dynamic program covers all required cells by such intervals, separated by empty squares; optional cells may remain empty. Failure is conclusive. Different rows are even allowed unlimited independent tile supplies, making this a relaxation of any real completion.

Four of the twenty patterns force the row

$$CO.Q \dots D \dots G \dots A,$$

which admits no such completion. The other sixteen leave an old X at B1, with A1 and C1 empty before the move. That tile must connect downward: it has no possible upward or horizontal neighbour. Enumerate every NWL23 word of length 2–15 beginning with X, respecting the existing letters, and repeat the permissive row-completion test. None succeeds. These two completion-safe checks reproduce the audit's split of 4 + 16 and eliminate every score at least 1724.

The attaining six-tile play places O, X, P, B, Z, E at A1, B1, D1, H1, L1, O1 to make OXYPHENBUTAZONE. The main word scores 1458; its crosses score 69 + 11 + 34 + 57 + 31 + 63 = 265, giving 1723. A checked 22-move setup uses 85 tiles with one non-scoring blank. Consequently the reproduced NWL23 vector through six tiles is

$$(M_1, \dots, M_6) = (213, 459, 882, 1130, 1401, 1723).$$

12. SEVEN TILES: BUYING AN UPPER BOUND WITH LETTER PRICES

The bingo adds exactly 50, so excluding scores above 1786 means excluding word-score sums at least 1737. Our fresh scan gives 87 main placements above this threshold. Their crosses cannot be optimized independently forever: the same two blanks must pay for every shortage.

Here is a useful bound with a short proof. Give each letter a any nonnegative rational price p_a . For cross option o in slot j , let s_{jo} be its all-real score and a_{jo} its vector of old-tile counts, excluding the already counted new tile. Let m be the main word's count vector and B the real bag. Put

$$z_j = \max_o (s_{jo} - p \cdot a_{jo}), \quad t = \max(0, \max_a (p_a - v(a))).$$

Lemma 9 (Resource prices). *Every scoring product through this main placement has pre-bingo score at most*

$$A + p \cdot (B - m) + \sum_j z_j + 2t.$$

If option o_j is chosen at some slots, subtract their nonnegative reduced costs $z_j - s_{jo_j} + p \cdot a_{jo_j}$ from this bound.

Proof. Let e_a be the number of blanks assigned to letter a in a feasible product. Its counts satisfy $m + \sum_j a_{j o_j} \leq B + e$ coordinatewise, with $\sum_a e_a \leq 2$. Each blank loses at least $v(a)$ because every scored physical square has coefficient at least one. Thus the score is at most

$$A + \sum_j (s_{j o_j} - p \cdot a_{j o_j}) + p \cdot (B - m) + \sum_a (p_a - v(a)) e_a.$$

Bound the last sum by $2t$ and each unchosen slot by z_j . This also proves the reduced-cost version. ■

A linear program proposes prices. We round them to multiples of $1/1024$ and recompute every maximum and inequality with integers. Any nonnegative prices give a valid certificate, even if the numerical optimizer did a poor job. Fixed-threshold enumeration retains all products, not merely the best few. This reproduces 3748 products across fourteen main placements in about 72 seconds on the development machine. Completion-safe row tests leave 255; mandatory hooks from isolated old top-row letters leave 34.

13. THE LAST 193 BOARDS All 34 residual moves form OXYPHENBUTAZONE across the top row. Delete their seven new tiles. Every remaining top-row component must have a downward connection somewhere: above it is the board edge, and beside it are squares that must remain empty until the final move. Choose one such connection from each component and enumerate every fitting dictionary word that could be its maximal vertical word. Enforce the common bag, but allow other support tiles later.

Incidental illegal rows must then be repaired. For each forced nonword, enumerate every possible final maximal word covering it; its interval cannot end next to an already forced letter. Retain compatible choices and their required additions under the common bag. Already legal runs need not be extended at this stage. This is complete as a reduction: in any genuine completion, select its actual words covering the forced nonwords and discard all other optional additions. One of the enumerated reduced boards remains as a subboard; discarded letters return to the available supply. Vertical nonwords at this stage are not grounds for rejection.

This shared-bag enumeration eliminates 33 patterns. The remaining pattern has new positions 1, 2, 4, 8, 11, 12, and 15, with crosses

OPACIFICATIONS, XEROSES, PREADJUSTING,
BRAINWASHING, AMELIORATIVE, ZOOGAMETE, EQUALITY.

These crosses contribute $69 + 14 + 26 + 63 + 17 + 31 + 60$ before the blank correction. Its reduced preboards form a set of exactly 193 distinct positions. Comparing sets of board coordinates is essential: two correct enumerators need not assign the same record numbers or serialize their records in the same order.

Every reduced board has three connected components, both blanks forced, and between two and six real tiles left after reserving the final seven tiles. What can these few letters connect?

Lemma 10 (Three-component connector). *Suppose a board has three components and some set of at most R new squares connects them. It contains a connecting subset that is the union of two simple paths from a suitable one of the components to the other two, with no path traversing a third old component.*

Proof. Contract each old component to a terminal vertex. Choose an inclusion-minimal connecting subset of new vertices, then a spanning tree, pruning nonterminal leaves. A tree connecting three terminals has either a terminal on the path between the other two, or one branching vertex with three terminal branches. In the first case choose the middle terminal as root; in the second choose any terminal. The two root-to-terminal paths cover the tree. Lift their new vertices back to squares. Each segment outside the old components is a simple path. If a path meets its target earlier through an extra adjacency, stop there; the union still connects all three terminals and uses a subset of the chosen new squares. ■

Enumerate such paths of length at most R , their unions of size at most R , and every assignment of the remaining letter multiset to those unions. Extra unused letters are not discarded: each row and each column may use the *entire* leftover multiset independently to repair its nonwords. This is more permissive than any common-board completion. A failed line therefore proves impossibility. The run checks 24,293 cores and 5,518,820 assignments. Row completion rejects 5,518,814; column completion rejects the last six. No 1787-point play remains.

Theorem 11 (North American maxima). *Under NWL23,*

$$(M_1, \dots, M_7) = (213, 459, 882, 1130, 1401, 1723, 1786).$$

Proof. The preceding exhaustive upper-bound reductions establish the exclusions. The first six attaining histories were described above. For seven, the supplementary atlas reproduces Lucassen's 1786 board. Its final words score $1458 + 69 + 11 + 34 + 13 + 57 + 31 + 63 + 50 = 1786$.

Its 97 tiles exceed the scope of Lemma 5, but a second player resolves the difficulty. Player B reserves the seven tiles needed for the last setup move, LADDERLIKE, and passes while A performs the first 23 setup moves. A's draw stream is those 83 setup tiles, followed by the seven record tiles and the three unused tiles. At that point A holds BENOPXZ. B plays LADDERLIKE and draws the last three tiles, so does not go out. A plays the record move and goes out. The independent history verifier checks all moves, alternating turns, blank assignments, racks, draws, and the bag. Endgame transfers are excluded from M_7 . ■

14. SIX TILES IN COLLINS: A TWO-LETTER BRIDGE For six tiles in Collins, a fresh scan at threshold 1722 covers 1,987,507,110 fitted main placements and retains 21. Their complete resource-constrained products leave 15,809 scoring patterns. Permissive row completion eliminates 12,457, and mandatory singleton hooks eliminate 3,345. Seven patterns remain. A shared-bag hook and row-completion enumeration eliminates all seven, establishing the upper bound

$$M_6(\text{CSW24}) \leq 1721.$$

Here an additional necessary test saves considerable work: for each forced nonword in a row, try every matching dictionary interval and allow it the entire remaining bag independently of all other rows and fragments. An actual completed word is one such interval and cannot use more than the remaining bag, including the two blanks. Thus failure safely prunes the partial board; success promises nothing about simultaneous completion. The checker retains the final-move tiles in the global supply count even though they are not yet on the board. Small exhaustive counterexample tests exercise this pruning, and rerunning the NWL23 hook stage still gives the same 193 reduced boards.

Proposition 12. *The six-tile maximum in CSW24 is 1721.*

Proof. For attainment, place O, Y, P, B, Z, E at A1, C1, D1, H1, L1, O1. The main word is again OXYPHENBUTAZONE, now with the X already present. Its score is 1458. The crosses, in order, are OPACIFYING, YARAKS, PREQUALIFIED, BRAINWASHING, ZOOGAMETES, and EJACULATIONS, giving

$$1458 + 63 + 13 + 29 + 63 + 32 + 63 = 1721.$$

The A at D6 is a blank and costs one point in PREQUALIFIED. The blank N at I4 is outside the scored words. The supplementary atlas gives the full 93-tile board.

A connected board alone would not finish the proof. The independent verifier re-plays 25 setup moves. They begin with SH through H8, grow it to RAINWASH, and use INTREAT and AQUILINE as bridges. Another particularly economical bridge is TOOT at L8–O8: its two middle letters connect the existing OOGAMETES and JACULATIONS columns. The last setup move extends PA to PACIFYING with seven tiles.

Player B reserves those seven tiles, passing while A plays the first 24 setup moves, which use 80 tiles. A's draw stream is those 80 tiles, then the six record tiles, then the seven unused tiles. Just before B's move, A holds BEEOPYZ. B plays PACIFYING and draws the six tiles left in the bag; A then plays the six-tile record and retains one E. No player has gone out prematurely. Every setup move scores positively, so the intervening passes do not terminate play. This is a full game-history certificate, not just a simultaneous word arrangement. ■

The two dictionaries therefore reverse order at six tiles. Compared with the NWL23 construction, the main score is unchanged. The six cross differences are $-6, +2, -5, +6, +1, 0$, summing to -2 . In particular, CSW24's extra vocabulary cannot replace the missing OPACIFICATIONS without a loss. Cardinality alone says nothing about inclusion of lexicons, much less monotonicity of their maxima.

15. SEVEN TILES IN COLLINS: THE PREDECESSOR MAKES THE DIFFERENCE

Proposition 13. *The seven-tile maximum in CSW24 is 1787.*

Proof. The complete scan at pre-bingo threshold 1738 covers 2,038,152,480 fitted placements and retains 168. Integer resource certificates and complete product enumeration leave 420,538 scoring patterns. Permissive row completion excludes 408,785; mandatory singleton hooks exclude 11,731 more; thirteen force the centre empty. Only nine remain, all scoring 1788 after the bingo. The shared-bag hook search eliminates all nine, proving the upper bound 1787. A separate streaming checker reconstructs every product's score and blank penalty and repeats the completion tests. Streaming changes storage, not the enumerated set or the pruning conditions.

For attainment, play O, X, P, B, A, Z, E at A1, B1, D1, H1, K1, L1, O1, again making OXYPHENBUTAZONE. The crosses, in order, are OPACIFYING, XED, PREQUALIFIED, BRAINWASHING, AMELIORATIVE, ZOOGAMETES, and EJACULATIONS. Figure 2 shows the construction. Its score is

$$1458 + 63 + 11 + 30 + 63 + 17 + 32 + 63 + 50 = 1787.$$

The only blanks are an A at C2 and an M at F3, both outside the scored words. They lose no points on the record move.

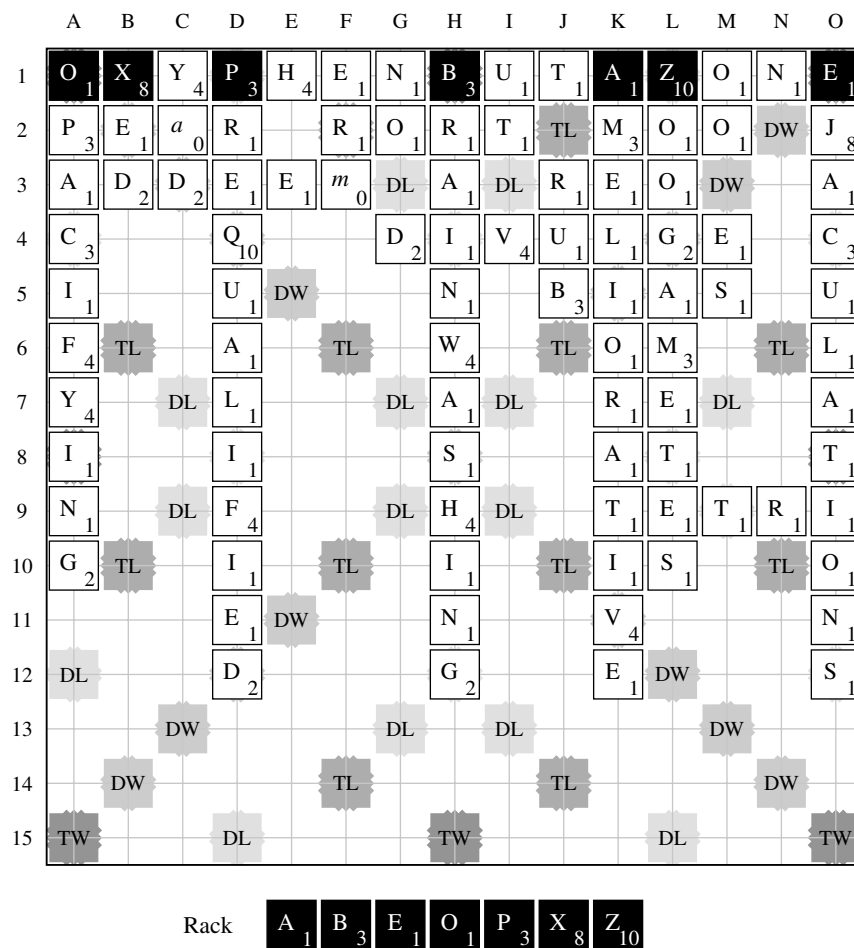


Figure 2. A 1787-point CSW24 construction. The rack is ABEOPXZ; black tiles are newly played. Both italic blanks, at C2 and F3, are auxiliary. The short bridge TETRI connects the right-hand column to the rest of the pre-final board.

The forward verifier checks 27 setup moves. Start with SH through H8, extend to RAINWASH, and use DIVULGE to reach the future OOGAMETE column. On the left, DEEM becomes ADDEEM; the short hook YAD attaches the old Y. The bridge TETRI at K9–O9 joins the right-hand component: its middle T, R are the two letters missing from the reduced board. Grow IONS to JACULATIONS on that bridge, then finish the remaining columns. Every intermediate word, score, and physical tile count is checked. The final board uses 96 tiles.

Player B reserves the seven tiles CFGIINY for the last setup move, PACIFYING. Player A's stream consists of the 82 tiles needed for the first 26 setup moves, then ABEOPXZ, then the four unused tiles. B passes between A's positive-scoring setup moves. Just before B's move, A has the record rack. B plays seven tiles and draws EKLW, so does not go out. A then plays the record and goes out. The explicit alternating rack/draw certificate verifies every step; endgame transfers are excluded from the claimed score. ■

The one-point gain over NWL23 is not explained by a more valuable main word:

both main words score 1458, and both receive the same bingo. Relative to the North American construction, the cross changes contribute $-6, 0, -4, +6, +1, 0$, and replacing the 13-point NARROWING cross by 17-point AMELIORATIVE adds four. Their sum is one. The new tile shifts from G1's N to K1's A; neither square has a premium.

More strikingly, all eight words scored by the Collins move belong to NWL23 too. But JACULATIONS, the word that must precede EJACULATIONS, does not. The support words ADDEEM, OO, and YAD are also Collins-only. Thus the final score table, even checked against both dictionaries, cannot certify a North American move. The deletion lemma detects the decisive obstruction before any board search.

Table 3. Exact ordinary move maxima. The difference is CSW24 minus NWL23; seven tiles include the 50-point bingo.

k	NWL23	CSW24	Difference
1	213	225	12
2	459	483	24
3	882	912	30
4	1130	1349	219
5	1401	1478	77
6	1723	1721	-2
7	1786	1787	1

16. REPRODUCIBILITY AND LIMITS OF THE EVIDENCE The accompanying source package contains programs, small certificate fixtures, constructions, tests, and version-matched input hashes. The top-level driver regenerates all fourteen claims from the dictionary files, rather than trusting stored verdicts. Separate checkers verify each attaining history. The upper-bound code has no SAT dependency. The numerical theorem remains computer-assisted: the finite dictionaries and their exhaustive treatment are part of the evidence, not a short dictionary-free inequality. This is not a Lean or Coq formalization.

The single-tile checker reruns both dictionaries. The common C++ scanner accepts k and a dictionary as arguments. The two-tile product enumerator retains every record at a fixed threshold, never just the best few. The NWL23 C++ exclusions for $k = 3, 4, 5$ require that no threshold record exist; their record display cap therefore cannot discard a successful exclusion counterexample. The NWL23 six-tile repair retains all twenty products and certifies their completion obstructions. The obsolete partial-board legality filter is not used to establish any theorem in this manuscript.

Board figures use the Scrabble TikZ package with final-size typography and monochrome-safe premium labels [11]. Standalone sources, vector PDFs, and EPS files accompany the manuscript. Research and code preparation used an AI assistant; responsibility for every mathematical claim remains with the authors. The public software is MIT-licensed; journal and TeX dependencies are fetched separately from hash-pinned official archives. The source package excludes licensed dictionary exports and private audit conversations, and explains how reviewers can supply version-matched dictionaries. Anonymous review requires a metadata-clean snapshot without identifying repository links.

17. HOW EXCEPTIONAL ARE THESE SCORES? There is no canonical uniform distribution on “all possible Scrabble plays.” A tournament sample measures human choices under competition; a sample of scoring patterns includes impossible

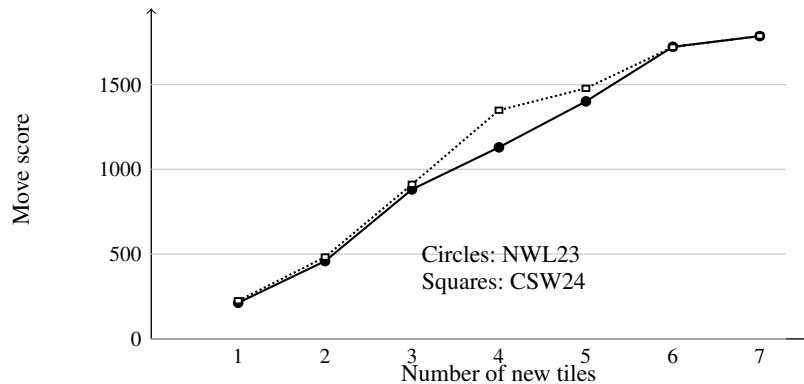


Figure 3. The two exact record curves. Their markers nearly coincide at six and seven tiles, where the differences are two points and one point respectively; Table 3 gives the exact values. The seven-tile step includes the 50-point bingo.

boards. Annotated games and statistics by tiles played are available through Cross-Tables and the RandomRacer project [12], but they are not used for Figure 4. To make the illustrative comparison completely reproducible, we instead specify a modest cooperative-game model.

The opening proposal chooses uniformly a dictionary word of length 2–7 and a horizontal position covering H8. Later proposals choose an occupied square uniformly, a matching dictionary word–letter occurrence uniformly among words of length 2–15, and an orientation uniformly. Accept the first proposal giving a legal positive-scoring move with one to seven new tiles and at most 86 physical tiles on the board. A real tile is used whenever available; an exhausted letter supply forces a permanent blank. Infeasible proposals are rejected. Stop after 24 accepted moves or after 3000 consecutive unsuccessful proposals. Racks are realized by Lemma 5; they are not independent random competition racks.

We generated 1000 independent-model games per dictionary, using deterministic per-game seeds derived by SHA-256 from master seed 20260910, the dictionary name, and the game index. Both fixed-count runs finished before their time limits; no time-censored partial sample is used. From each game containing a post-opening k -tile move, choose one such move uniformly. Thus N_k counts qualifying games, not correlated turns, and the plotted law is conditional on a game containing such a move. All 2000 histories, physical bags, rack streams, and sampled scores were independently replayed.

Write $\hat{F}_k$ for a plotted empirical distribution and F_k for its model law. Conditional on N_k , the Dvoretzky–Kiefer–Wolfowitz–Massart bound [13] and a union bound give simultaneous 95% bands for all fourteen curves:

$$\max(0, \hat{F}_k - \epsilon_k) \leq F_k \leq \min(1, \hat{F}_k + \epsilon_k), \quad \epsilon_k = \sqrt{\frac{\log(560)}{2N_k}}.$$

As usual for a seeded simulation, this uncertainty statement concerns the ideal independent sampling model, not a claim of randomness of the fixed seed. The panels report N_k ; the exact summary data also accompany the source. Band half-widths range from 0.057 to 0.152. The one-tile estimates are notably less precise: their half-widths are 0.138 for NWL23 and 0.152 for CSW24. The very visible seven-tile shift largely reflects the 50-point bonus; neither the observed score frequencies nor the largest sam-

pled value, 239, estimates an extremal score. No conclusion about tournament scoring frequencies follows from this experiment.

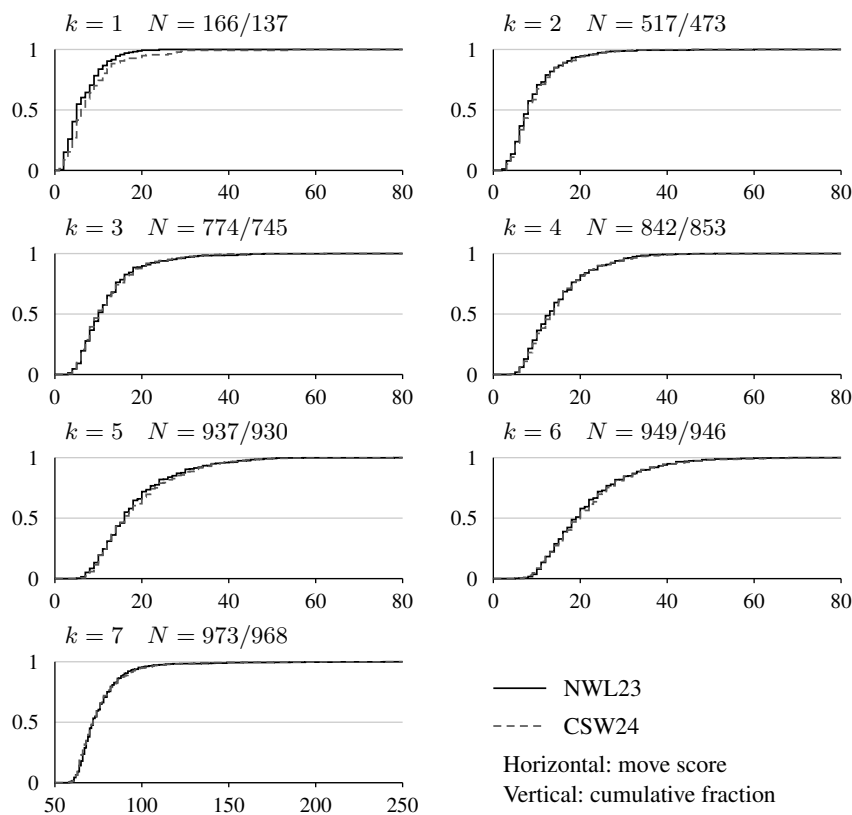


Figure 4. Empirical score distributions under the specified cooperative-game model, not tournament statistics or a uniform sample of legal plays. Each panel gives NWL23/CSW24 sample sizes. The seven-tile panel uses a different horizontal scale so that its entire tail is shown.

18. WHAT MAKES A RECORD POSSIBLE The largest words are only the beginning. Premiums explain the scale, but a repeated letter can exhaust the blanks, a missing predecessor can invalidate a cross, and a last bridge can exhaust the bag. A record needs all three objects: scoring words, a legal board, and an actual game. Our upper bounds relax these requirements explicitly; the attaining histories restore them.

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